

HW2 – Part 3

Preparation

Please make sure you have read Lecture 6

Please read Textbook chapter 16, at least the portions covered by the lectures.

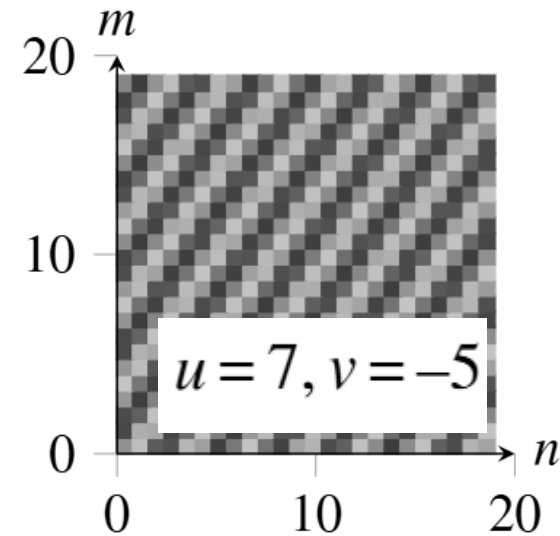
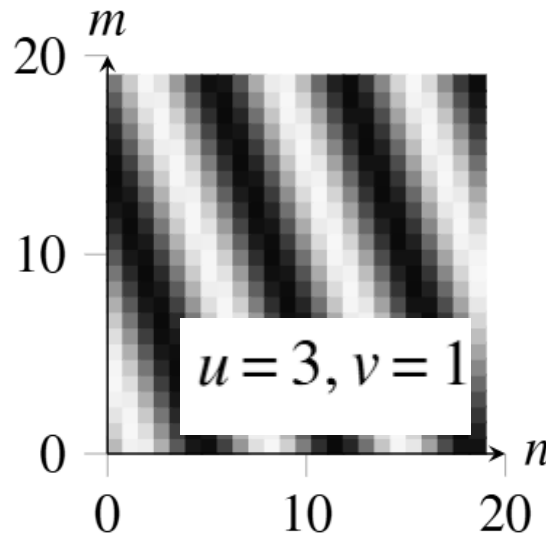
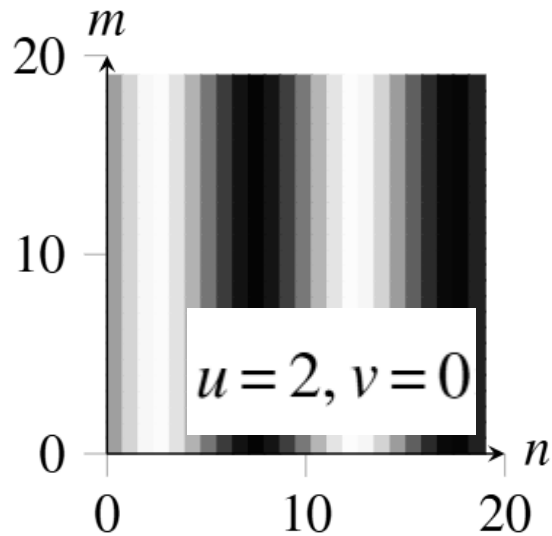
The notations in this slide deck may be a bit different from the lecture slides!
The goal is to make implementation easier.

Caution

- Please do not import packages (like scikit learn) that are not listed in the provided code.
- In this homework, you are NOT allowed to use NumPy's or other Python libraries' built-in **convolution, DFT, IDFT, and filter functions**. If you use them, you will get 0 points for the entire assignment.

Question 1: Periodic function

- Let's consider a square image
- $l[n, m] = \cos\left(2\pi\left(\frac{un}{N} + \frac{vm}{N}\right) + \theta\right)$, where θ is the phase

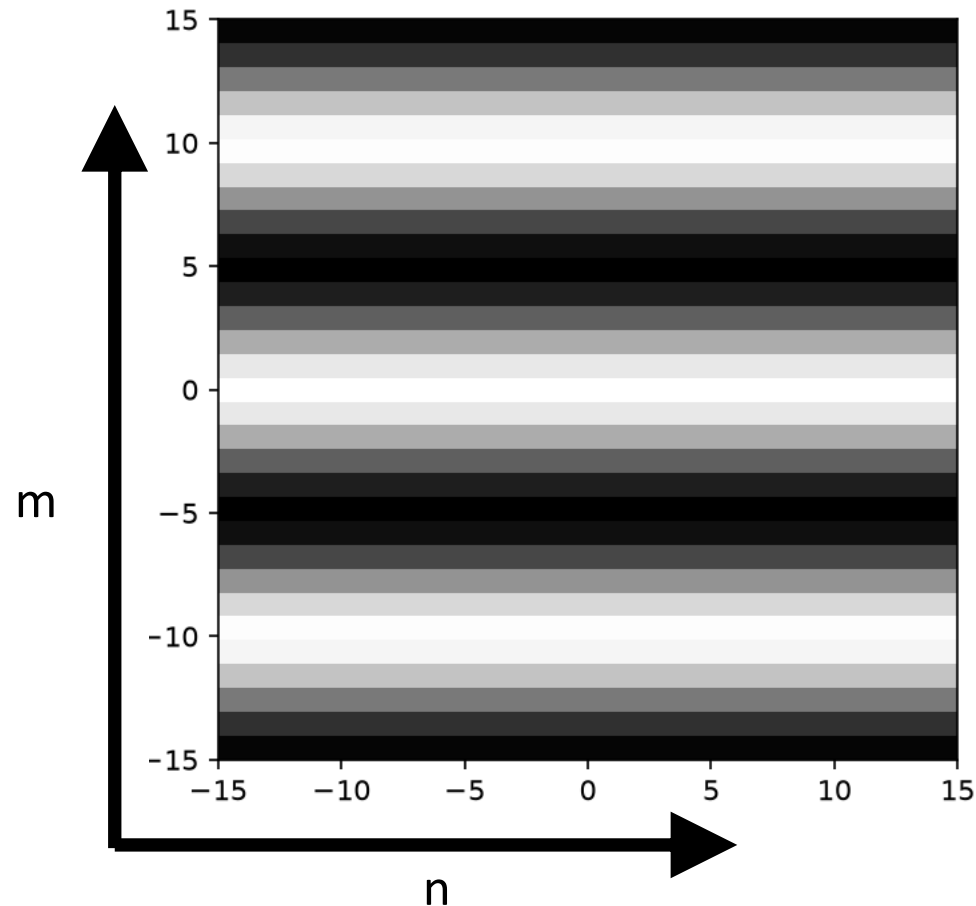


Question 1: Periodic function

- $l[n, m] = \cos \left(2\pi \left(\frac{un}{N} + \frac{vm}{N} \right) + \theta \right)$, where θ is the phase
- In this homework, please assume N is an odd number
- For n , consider the range $-(N-1)/2 : (N-1)/2$
- For m , consider the range $-(N-1)/2 : (N-1)/2$

Question 1: Periodic function

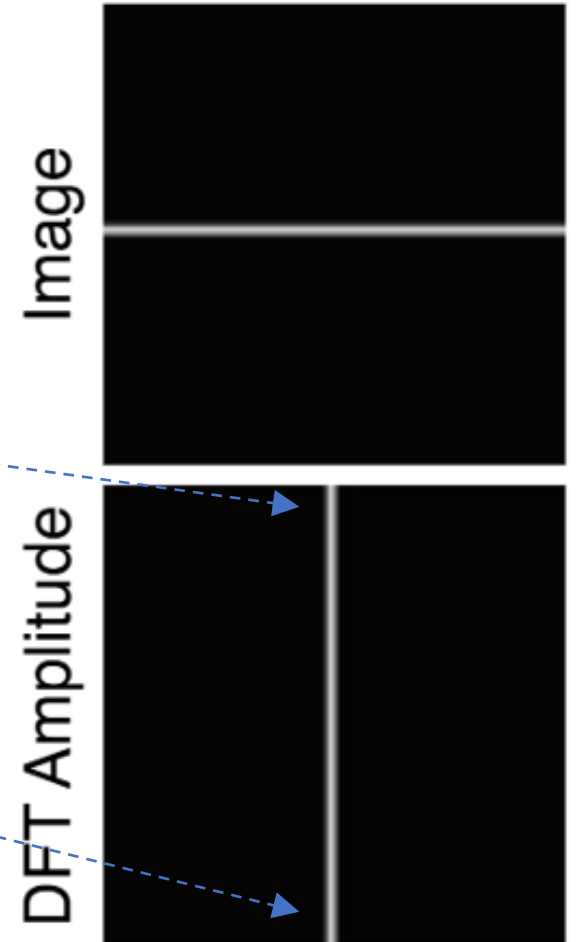
- Let $N = 31$, $u = 0$, $v = 3$, and $\theta = 0$, you will generate a 31-by-31 matrix (image)



Other experiments

- In Lecture 6, we see this example
- Now let's reproduce it by

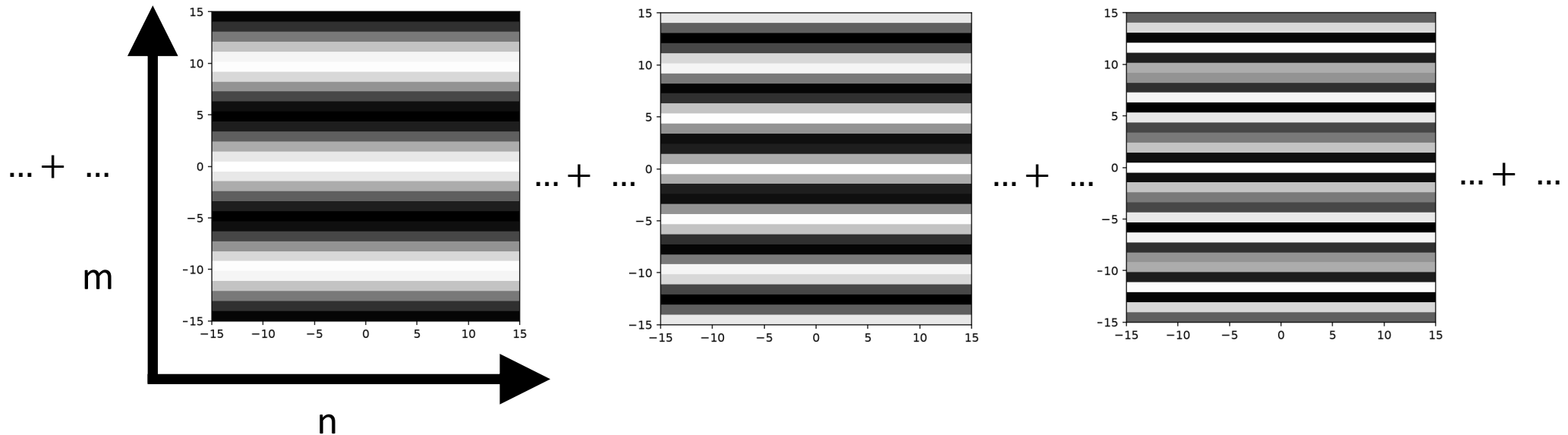
$$l[n, m] = \sum_{v=-(N-1)/2}^{v=(N-1)/2} \cos \left(2\pi \left(\frac{\textcolor{red}{0}n}{N} + \frac{\textcolor{blue}{v}m}{N} \right) \right)$$



Other experiments

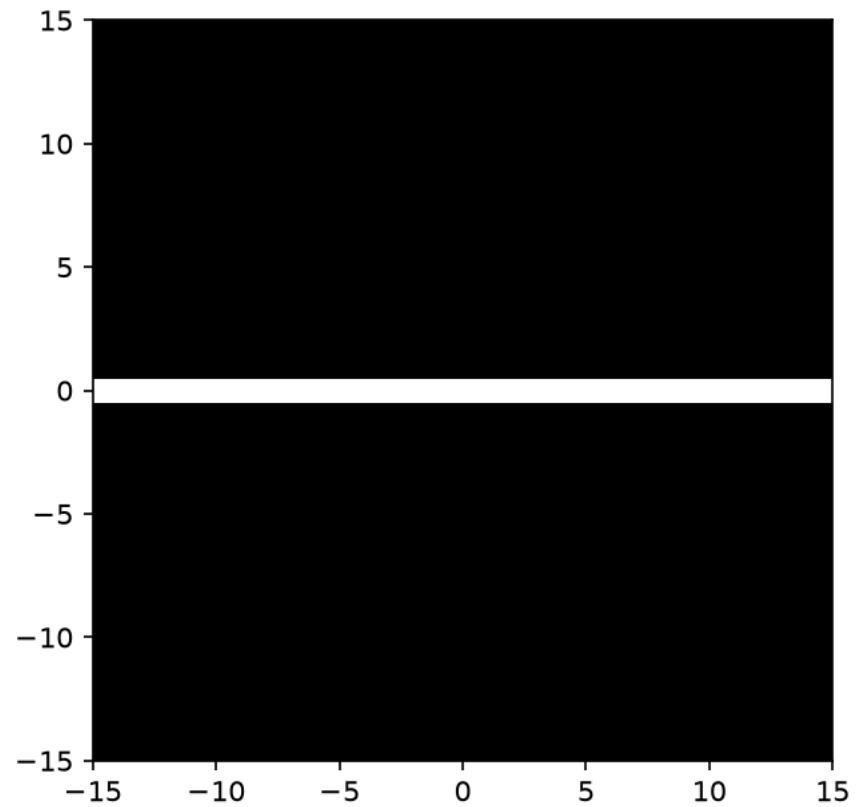
$$l[n, m] = \sum_{v=-(N-1)/2}^{v=(N-1)/2} \cos \left(2\pi \left(\frac{0n}{N} + \frac{vm}{N} \right) \right)$$

means



Other experiments

- You will get



Other experiments

- How about $l[n, m] = \sum_{v=-(N-1)/2}^{v=(N-1)/2} \cos \left(2\pi \left(\frac{0n}{N} + \frac{vm}{N} \right) + \theta \right),$
- where θ is a random number for each v value?